MUMBAI UNIVERSITY

B.E. / T.E. INFORMATION TECHNOLOGY – SEMESTER VII

MASTER TOP 10 QUESTIONS PER SUBJECT

SUBJECT 1: ARTIFICIAL INTELLIGENCE & DATA SCIENCE – II (AI & DS II)

1. Explain various metrics for evaluating the classifier performance. Discuss any two cross validation methods.

   Numerical Part: Calculate Accuracy, precision, recall, sensitivity and specificity for the given confusion matrix example.

2. Explain the architecture and components of CNN in detail and state its application also.
3. Briefly explain Data science for Multimodal applications. Discuss how different data modalities (text, audio, image, and video) are integrated to improve performance.
4. How does Bayes Theorem determine the probability of an event with uncertain knowledge? Give one example of it. (Includes numericals on medical diagnosis probability / card posterior probability).
5. Using Mamdani fuzzy model design a fuzzy logic controller to determine the wash time of domestic washing machine. Assume the inputs are dirt and grease on clothes. Use three descriptors for each input variable and five descriptor for output variables. Derive necessary membership functions and required fuzzy rules for the application.
6. How bagging and boosting handle bias-variance trade-off differently, and analyze their effectiveness in dealing with noisy data and overfitting. Explain with algorithmic such as Random forest and AdaBoost.
7. Define Cognitive computing. Draw a neat diagram of components of the cognitive system and explain the components also List and explain the design principle of Cognitive system. What is cognitive computing, and how does it differ from traditional computing?
8. Explain Mc-Culloch pits neuron with an example. Implement AND, XOR function using Mc-Culloch Pitts model (use binary data representation).
9. Define Defuzzification. Discuss any two methods of defuzzification with formulas and advantages.
10. What is the Bayesian Belief Network? Explain Interfacing in detail with example.

SUBJECT 2: INTERNET OF EVERYTHING (IOE)

1. Draw and explain the architecture of MQTT with its advantages, topic hierarchy, publish-subscribe broker, and Quality of Service (QoS 0, 1, 2) levels with packet exchange diagrams.
2. Compare and contrast Edge, Fog, and Cloud Computing. Explain their usage with an example and illustrate the 3-tier hierarchy between Edge, Fog, and Cloud.
3. Explain the working of Constraint Application Layer Protocol (CoAP) with its 2-layer architecture, message types (CON, NON, ACK, RST), and RESTful client-server model over UDP.
4. Explain effective Data Visualization methods with suitable examples (line charts, heat maps, real-time gauges, scatter plots, dashboards) and their principles in IoT.
5. Explain the architecture of LoRaWAN with its major characteristics, network components (End Nodes, Gateways, Network Server, Application Server), and device classes (Class A, B, C).
6. Explain the Smart Object with diagram and describe its characteristics, components (sensors, processing unit, memory, communication, actuators), and operating constraints.
7. Draw and explain the architecture of Advanced Message Queuing Protocol (AMQP) with advantages (Producers, Exchange types, Binding keys, Message Queues, Consumers).
8. Explain the term in detail: Internet of Behaviour (IoB) with its definition, data sources, behavioral analytics lifecycle, applications, and ethical privacy challenges.
9. Explain the architecture of RFID with its major applications, working principle, and differences between Passive RFID, Active RFID, and Bluetooth.
10. Explain the importance of IoT data analytics. Compare and contrast IoT data analytics and IoT network analytics with respect to metrics, focus areas, and outputs.

SUBJECT 3: SOFTWARE TESTING & QUALITY ASSURANCE (STQA)

1. Consider a given C program. Draw the DD-graph (Decision-to-Decision graph), calculate cyclomatic complexity using all three methods (Edges/Nodes formula, Predicate node formula, Enclosed regions formula), list out independent basis paths, and design test cases.
2. Explain Agile Testing Life Cycle (ATLC), Agile testing quadrants (Q1 to Q4), principles, and major challenges faced in Agile testing sprints.
3. What is Mutation Testing? Explain primary mutants, secondary mutants, mutant kill ratio, equivalent mutants, and write the formula for Mutation Score (Adequacy).
4. Explain criteria for selecting a test automation tool. Describe Selenium WebDriver architecture with a neat diagram and its advantages over Selenium RC.

5. Explain the structure of a testing group and test management elements (test planning, scheduling, staffing, and test milestone tracking) in a software organization.
6. Explain Boundary Value Analysis (BVA) and Equivalence Class Partitioning (ECP). Design robust and normal boundary test cases for a given input specification.
7. Explain Defect Spoilage. Calculate defect spoilage metric value given a defect discovery and lifecycle stage weightage table.
8. Explain McCall's Software Quality Factors with diagram (Product Operation, Product Revision, and Product Transition categories and their 11 sub-factors).
9. Draw and explain the Bug / Defect Life Cycle with a state transition diagram showing all states (New, Assigned, Open, Fixed, Retest, Verified, Closed, Reopened, Deferred).
10. Explain basic testing terminology: Differentiate between Error, Fault, Bug, and Failure with real-world software examples.

SUBJECT 4: INFORMATION RETRIEVAL SYSTEM (IRS)

1. Describe the process of creating an inverted index with an example. How can this process be optimized using block addressing?
2. Explain User Relevance Feedback. Discuss how a user relevance feedback model (Rocchio Algorithm) can be applied to improve search results in a search recommendation engine.
3. Explain Keyword-based querying: Phrase queries, Proximity queries, and Wildcard queries with suitable examples.
4. Discuss Natural Language Modelling issues (Polysemy, Synonymy, Morphological variations) and solutions in Information Retrieval systems.
5. What is Multimedia Information Retrieval (MIR)? Explain Video Information Retrieval (VIR) and Image Information Retrieval (IIR) indexing and search techniques.
6. Discuss Human-Computer Interaction (HCI) design principles and result visualization (TileBars, Scatter/Gather) in Information Retrieval.
7. Explain the objectives of an Information Retrieval System and describe the complete retrieval process pipeline with a neat diagram.
8. Explain Vector Space Model and Cosine Similarity Ranking with mathematical formulas. Calculate tf-idf weights, document lengths, and rank documents for a given query and collection.
9. Explain Search Engine Architecture: Web Crawling (Crawler/Spider components, Polite crawling, Robots.txt protocol), Indexing, and Query Processing.
10. Explain Suffix Data Structures: Suffix Trie, Suffix Tree, and Suffix Array with construction diagrams and complexity comparison.

SUBJECT 5: CYBER SECURITY & LAWS (CS&L)

1. What is buffer overflow? Explain different buffer overflow attacks (stack-based and heap-based). Also explain how to mitigate buffer overflow attacks using ASLR, DEP/NX-bit, and Stack Canaries.
2. Explain what is Information Security Standard and write key IT requirements for SOX (Sarbanes-Oxley Act) and HIPAA (Health Insurance Portability and Accountability Act) compliance in detail.
3. Define E-Contract. Explain the different types of E-Contracts (Click-wrap, Shrink-wrap, and Browse-wrap) and their legal enforceability under the Indian IT Act 2000.
4. What is cyber crime? How are cybercrimes classified (against individuals, property, and organization)? Explain cyber criminals categories and their motivations.
5. What is SQL injection? Explain the steps of SQL injection with an authentication bypass example. Explain types of SQL injection and countermeasures/prevention methods.
6. What is digital evidence? Explain rules of digital evidence, evidence collection, chain of custody, and admissibility under Section 65B of Indian Evidence Act.
7. Explain different phases of planning cyber attacks by criminals (Reconnaissance, Scanning, Gaining Access, Maintaining Access, Covering Tracks).
8. Explain Denial of Service (DoS) and Distributed Denial of Service (DDoS) attack mechanisms in detail with amplification attacks and countermeasures.
9. What are different Security Risks for Organizations in cyberspace? Outline the challenges for securing data from a business perspective.
10. Explain cyber cafe illegal activities, security issues, vulnerabilities, and safety precautions required by cyber cafe owners.

END OF MUSA MASTER TOP 10 QUESTIONS PER SUBJECT

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